

# Cardiovascular Disease Prediction Using Machine Learning

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## Abstract

This paper presents three different machine learning models and their results based on a cardiovascular dataset which consists of medical data given by patients and recorded by medical personnel. Logistic Regression, Random Forest and Artificial Neural Networks were applied and their results have been analyzed and compared by certain metrics. The goal of this paper is to find the most optimal model and the model which will produce the least False Positives which are the most important factor for models applied in the medical field.

## 1 Introduction

Cardiovascular diseases are among the leading causes of death worldwide, making early detection and prevention essential for improving patient outcomes. With the increasing availability of medical data, machine learning techniques have become valuable tools for assisting healthcare professionals in identifying individuals at risk of developing cardiovascular conditions in the future.

The three models which were tested are Logistic Regression, Random Forest, and Artificial Neural Networks. The models are trained and evaluated using a dataset containing patient medical information and clinical measurements. Their performance is compared using several evaluation metrics, including precision, recall, F1-score, and ROC-AUC. The objective of this study is to determine which model provides the most reliable predictions and is most suitable for supporting cardiovascular disease diagnosis.

## 2 Dataset

The dataset used in this study is the Cardiovascular Disease Dataset, which contains medical records collected from patients during routine clinical examinations. The dataset consists of approximately 70,000 observations and includes demographic, physical, and clinical measurements that may be associated with cardiovascular disease.

The available features include age, gender, height, weight, systolic blood pressure (ap\_hi), diastolic blood pressure (ap\_lo), cholesterol level, glucose level, smoking status, alcohol consumption, and physical activity. The target variable, cardio, is binary, where a value of 1 indicates the presence of cardiovascular disease and a value of 0 indicates its absence.

## 3 Data Preprocessing

Before training the machine learning models, several preprocessing steps were performed. Age values originally stored in days were converted to years, while records containing unrealistic values for height, weight, and blood pressure were

removed. Additionally, entries where the systolic blood pressure was lower than the diastolic blood pressure were excluded to ensure data quality and consistency. From 70 000 records only 68 118 were left.

The resulting dataset was then used for exploratory data analysis, feature evaluation, and model training.

The following values were received after the data cleaning.

Table 1: Descriptive Statistics (Part 1)

| Statistic | age      | gender   | height   | weight   | ap_hi    |
|-----------|----------|----------|----------|----------|----------|
| count     | 68188.00 | 68188.00 | 68188.00 | 68188.00 | 68188.00 |
| mean      | 53.33    | 1.35     | 164.51   | 74.19    | 126.72   |
| std       | 6.76     | 0.48     | 7.77     | 13.95    | 16.65    |
| min       | 29.60    | 1.00     | 140.00   | 45.00    | 60.00    |
| 25%       | 48.40    | 1.00     | 159.00   | 65.00    | 120.00   |
| 50%       | 54.00    | 1.00     | 165.00   | 72.00    | 120.00   |
| 75%       | 58.40    | 2.00     | 170.00   | 82.00    | 140.00   |
| max       | 65.00    | 2.00     | 207.00   | 150.00   | 240.00   |

Table 2: Descriptive Statistics (Part 2)

| Statistic | ap_lo    | cholesterol | gluc     | smoke    | alco     | active   |
|-----------|----------|-------------|----------|----------|----------|----------|
| count     | 68188.00 | 68188.00    | 68188.00 | 68188.00 | 68188.00 | 68188.00 |
| mean      | 81.33    | 1.37        | 1.23     | 0.09     | 0.05     | 0.8      |
| std       | 9.42     | 0.68        | 0.57     | 0.28     | 0.22     | 0.4      |
| min       | 40.00    | 1.00        | 1.00     | 0.00     | 0.00     | 0.0      |
| 25%       | 80.00    | 1.00        | 1.00     | 0.00     | 0.00     | 1.0      |
| 50%       | 80.00    | 1.00        | 1.00     | 0.00     | 0.00     | 1.0      |
| 75%       | 90.00    | 2.00        | 1.00     | 0.00     | 0.00     | 1.0      |
| max       | 180.00   | 3.00        | 3.00     | 1.00     | 1.00     | 1.0      |

## Feature Correlation Conclusions

To show the feature correlation the Pearson correlation coefficient was calculated and the following results were observed:

- There is a **strong positive correlation** between systolic (ap\_hi) and diastolic (ap\_lo) blood pressure ( $r \approx 0.73$ ), which is expected since both measure related physiological processes.
- **Blood pressure shows the strongest correlation with cardiovascular disease:**
  - ap\_hi (systolic):  $r = 0.43$
  - ap\_lo (diastolic):  $r = 0.34$

This confirms that blood pressure is one of the most important risk factors in the dataset.

- **Height and gender** are moderately correlated ( $r = 0.52$ ), reflecting natural physiological differences between sexes.
- **Height and weight** show a moderate correlation ( $r \approx 0.30$ ), reflecting natural body composition relationships.
- **Cholesterol and glucose** levels are moderately correlated ( $r \approx 0.45$ ), suggesting a link between metabolic health factors.
- **Age** shows a moderate positive correlation with cardiovascular disease ( $r = 0.24$ ), confirming that older patients carry a higher risk.
- **Lifestyle features** (smoke, alco, active) have very weak correlations with the target variable ( $|r| \leq 0.04$ ). This may indicate:
  - Non-linear relationships, or
  - Interactions not captured by simple correlation analysis.
- **Gender** shows almost no correlation with cardiovascular disease ( $r = 0.01$ ), meaning it is not a strong standalone predictor in this dataset.

## 4 Logistic Regression

### 4.1 Model Setup

A Logistic Regression model was trained to predict the presence of cardiovascular disease (cardio). The dataset was split into 80% training and 20% test sets using stratified sampling (random\_state=50) to preserve class balance. All features were standardised using StandardScaler prior to training to ensure that coefficients are comparable across features.

### 4.2 Feature Coefficients

Table 3 presents the learned coefficients sorted by absolute magnitude. A positive coefficient indicates that higher values of a feature increase the predicted probability of cardiovascular disease, while a negative coefficient indicates a protective or inverse relationship.

**Table 3: Logistic Regression Coefficients (sorted by absolute value)**

| Feature     | Coefficient |
|-------------|-------------|
| ap_hi       | 0.9318      |
| age         | 0.3374      |
| cholesterol | 0.3241      |
| weight      | 0.1589      |
| ap_lo       | 0.1031      |
| active      | -0.0955     |
| gluc        | -0.0658     |
| smoke       | -0.0421     |
| alco        | -0.0415     |
| height      | -0.0348     |
| gender      | -0.0065     |

The dominant predictor is systolic blood pressure (ap\_hi,  $\beta = 0.932$ ), followed by age ( $\beta = 0.337$ ) and cholesterol ( $\beta = 0.324$ ). These findings are consistent with the correlation analysis presented earlier. Physical activity (active,  $\beta = -0.096$ ) is the strongest negatively associated feature, suggesting a mild protective effect.

### 4.3 Classification Report

**Table 4: Classification Report – Logistic Regression**

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| No Disease   | 0.71      | 0.79   | 0.75     | 6878    |
| Disease      | 0.76      | 0.67   | 0.71     | 6760    |
| Accuracy     |           |        | 0.73     | 13638   |
| Macro avg    | 0.73      | 0.73   | 0.73     | 13638   |
| Weighted avg | 0.73      | 0.73   | 0.73     | 13638   |

The model achieves an overall accuracy of **73%**. Precision and recall are reasonably balanced across both classes, with the model performing slightly better at identifying healthy patients (recall = 0.79) than diseased ones (recall = 0.67). The ROC-AUC score of **0.797** indicates good discriminative ability, meaning the model correctly ranks a randomly chosen diseased patient above a healthy one approximately 80% of the time.

### 4.4 Confusion Matrix

From the confusion matrix the following counts were obtained on the test set ( $n = 13,638$ ):

- **True Negatives (TN):** 5456 — correctly predicted No Disease
- **False Positives (FP):** 1422 — healthy patients incorrectly flagged as Disease
- **False Negatives (FN):** 2243 — diseased patients missed by the model
- **True Positives (TP):** 4517 — correctly predicted Disease

The false negative count (2243) is notably higher than the false positive count (1422), meaning the model tends to under-predict disease. In a clinical context this is the more costly type of error, as missed diagnoses carry greater risk than false alarms.

### 4.5 Summary

Logistic Regression provides a solid and interpretable baseline with an accuracy of 73% and a ROC-AUC of 0.80. The model confirms that ap\_hi, age, and cholesterol are the strongest linear predictors of cardiovascular disease. However, the relatively high false negative rate suggests that a more complex, non-linear model may better capture the underlying patterns in the data.

## 5 Random Forest

A Random Forest classifier was trained using an ensemble of 100 decision trees with a maximum depth of 10, a random state of 50, and no parallelisation ( $n\_jobs=1$ ). The model was evaluated on the held-out test set using accuracy, F1-score, ROC-AUC, and a confusion matrix.

### 5.1 Model Performance

The Random Forest achieved an overall accuracy of **0.74** and a macro-averaged F1-score of **0.74**, marginally outperforming the Logistic Regression baseline (accuracy 0.73, F1 0.73) across all metrics. The ROC-AUC score of **0.8068** further confirms a modest but consistent improvement over the linear model ( $AUC = 0.796$ ), as illustrated in the ROC curve comparison.

The per-class breakdown revealed an asymmetry between the two classes. For the *No Disease* class, the model achieved a precision of 0.72 and a recall of 0.80, while for the *Disease* class precision was 0.77 with a lower recall of 0.68. This disparity indicates that the model is more conservative in predicting disease, which has direct clinical implications discussed in Section 5.3.

The confusion matrix confirms this behaviour: of 6,878 healthy patients, 5,527 were correctly classified (true negatives) and 1,351 were incorrectly flagged as diseased (false positives). Of 6,760 disease patients, 4,581 were correctly identified (true positives), but **2,179 were missed** (false negatives). In a medical screening context, these missed cases represent the most critical failure mode, motivating the threshold analysis presented below.

### 5.2 Feature Importance

Systolic blood pressure ( $ap\_hi$ ) was by far the dominant predictor, accounting for approximately 47% of the total importance. Diastolic blood pressure ( $ap\_lo$ ) ranked second at around 19%, followed by age at approximately 15%. Cholesterol contributed a moderate importance of roughly 10%, while weight and height provided minor contributions. Lifestyle features—physical activity ( $active$ ), smoking ( $smoke$ ), alcohol consumption ( $alco$ ), and gender—showed negligible predictive power (each below 2%).

These results are consistent with established medical literature: hypertension and advancing age are well-recognised primary risk factors for cardiovascular disease. The low importance of lifestyle variables is likely attributable to the self-reported nature of those features in the dataset, which introduces substantial measurement noise and reduces their discriminative signal.

### 5.3 Threshold Optimisation

By default, binary classifiers apply a decision threshold of 0.5 to the predicted probability. In a clinical screening context, however, false negatives (missed disease cases) are considerably more harmful than false positives (unnecessary follow-up), as an undetected condition may lead to delayed treatment and adverse outcomes. A false positive, while causing undue stress,

can be resolved through further examination. This asymmetry motivates a systematic evaluation of lower classification thresholds.

Table 5 reports accuracy, disease recall, disease F1-score, and ROC-AUC for thresholds ranging from 0.30 to 0.50. The AUC remains constant at 0.8068 across all thresholds, as it is a threshold-independent metric of overall discriminative ability.

**Table 5: Threshold comparison for the Random Forest classifier.**

| Threshold | Accuracy | Disease Recall | Disease F1 | AUC    |
|-----------|----------|----------------|------------|--------|
| 0.30      | 0.6873   | 0.8849         | 0.7372     | 0.8068 |
| 0.35      | 0.7116   | 0.8445         | 0.7438     | 0.8068 |
| 0.40      | 0.7305   | 0.7851         | 0.7428     | 0.8068 |
| 0.45      | 0.7366   | 0.7259         | 0.7321     | 0.8068 |
| 0.50      | 0.7412   | 0.6777         | 0.7219     | 0.8068 |

As the threshold decreases, disease recall rises substantially—from 0.6777 at the default threshold to 0.8849 at  $t = 0.30$ —reflecting that fewer disease cases are missed. This improvement comes at the cost of overall accuracy, which falls from 0.7412 to 0.6873 over the same range, as more healthy patients are incorrectly classified as diseased.

A threshold of **0.40** was selected as the optimal operating point. At this threshold, the number of false negatives was reduced by approximately 30% (from 2,179 to 1,453 cases) while overall accuracy remained competitive at 0.73 and disease F1-score improved from 0.72 to 0.74.

## 6 Artificial Neural Network

A baseline Artificial Neural Network (ANN) was trained using scikit-learn's `MLPClassifier` with two hidden layers of 64 and 32 neurons respectively, ReLU activation functions, a maximum of 500 iterations, and a fixed random state of 50. The input features were standardised prior to training using the same scaler fitted on the training set. The model was evaluated on the held-out test set using the same metrics applied to the preceding classifiers.

### 6.1 Model Performance

The baseline ANN achieved an overall accuracy of **0.73** and a macro-averaged F1-score of **0.73**, placing it on par with Logistic Regression and just below the Random Forest. The ROC-AUC score of **0.7945** situates the ANN between Logistic Regression (0.796) and Random Forest (0.805), confirming that despite its greater architectural complexity it did not achieve superior discriminative performance on this dataset.

The per-class classification report revealed near-symmetric behaviour across the two classes. For the *No Disease* class, the model attained a precision of 0.73 and a recall of 0.74. For the *Disease* class, precision was 0.73 with a recall of 0.72. The close agreement between macro-averaged and weighted-averaged metrics (both 0.73) reflects the near-balanced class distribution in the test set.

The confusion matrix shows that of 6,878 healthy patients, 5,122 were correctly classified as healthy (true negatives) and 1,756 were incorrectly flagged as diseased (false positives). Of 6,760 disease patients, 4,837 were correctly identified (true positives), while **1,923 were missed** (false negatives). Compared to the Random Forest at the default threshold (2,179 false negatives), the ANN produced fewer missed disease cases, though this came alongside a higher false positive count.

## 6.2 Training Convergence

I have displayed in the project the training loss curve over 200 iterations (plotted from the `loss_curve_` attribute, which records the loss at each iteration). The loss decreases rapidly in the first 50 iterations, dropping from approximately 0.570 to 0.540, before entering a prolonged phase of gradual improvement that continues to around iteration 200, where the curve begins to plateau near 0.525. The smooth and monotonically decreasing trajectory indicates stable convergence without signs of divergence or oscillation, suggesting that the learning rate and architecture were appropriate for this task.

## 6.3 Discussion

The ANN's failure to outperform the Random Forest is consistent with findings reported in the machine learning literature concerning structured tabular medical data. Ensemble tree-based methods, such as Random Forest, tend to match or surpass neural networks on datasets of this scale and dimensionality, particularly when the underlying feature relationships are relatively linear and the number of samples is in the tens of thousands rather than millions. The marginal gains of the ANN over Logistic Regression further support the observation, noted in the Random Forest analysis, that the cardiovascular disease prediction task on this dataset does not exhibit substantial non-linear complexity.

These results suggest that the additional computational cost and hyperparameter sensitivity of a neural network are not justified for this particular problem, and that a well-tuned Random Forest with threshold optimisation represents the more practical and interpretable modelling choice.

## 7 Conclusion and Future Work

Among the three models evaluated, the Random Forest classifier demonstrated the strongest overall performance and, following threshold optimisation, offered the most clinically suitable trade-off between sensitivity and specificity. Logistic Regression proved to be a competitive and interpretable baseline, while the Artificial Neural Network, despite its architectural complexity, did not provide meaningful gains over the simpler models.

In clinical practice, ensemble tree-based methods such as Random Forest and gradient boosting variants are widely used for cardiovascular risk prediction on structured tabular data, owing to their robustness, ability to capture non-linear feature interactions, and relative interpretability through feature importance scores. Logistic Regression remains common in

clinical decision support tools precisely because its coefficients are directly interpretable by medical professionals. Neural networks, while dominant in imaging and signal-based cardiac diagnostics, have shown limited advantage over tree-based methods on tabular clinical datasets, a finding well established in the literature.

The results obtained in this study are consistent with these expectations. Random Forest outperformed Logistic Regression, though only marginally, suggesting that the feature relationships in this dataset are predominantly linear in nature. The negligible gain from the ANN further supports this conclusion. The dominance of blood pressure, age, and cholesterol as predictive features also aligns with established cardiovascular risk frameworks such as the Framingham Heart Study, lending validity to the findings despite the modest overall performance.

Several avenues exist for improving upon these results. Gradient boosting methods such as XGBoost or LightGBM would be natural next steps, as they typically outperform standard Random Forests on tabular medical data. Finally, external validation on an independent patient cohort would be necessary before any of these models could be considered for real-world clinical deployment.